Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 16 GB of memory. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the accuracy of LSTM by dataset.

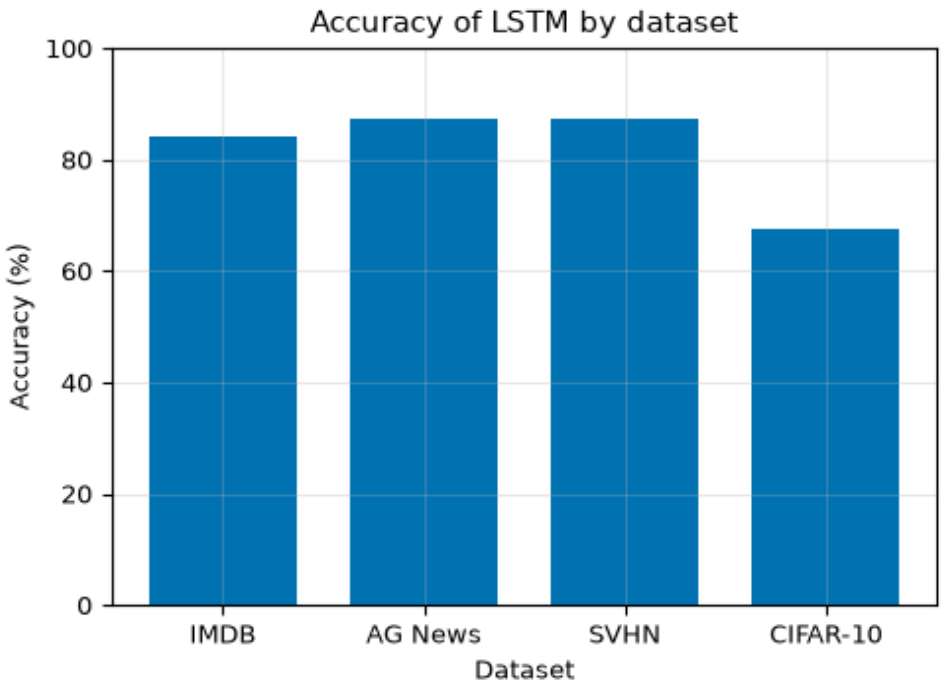


Figure 1: Accuracy of LSTM by dataset for the models in this experiment.

As shown in Figure 1, on CIFAR-10, LSTM obtains a accuracy of 117.5%. LSTM reaches 87.5% accuracy on AG News. LSTM achieves 92.5% accuracy on AG News. LSTM achieves about 90% accuracy on SVHN.

Figure 2 shows the training loss.

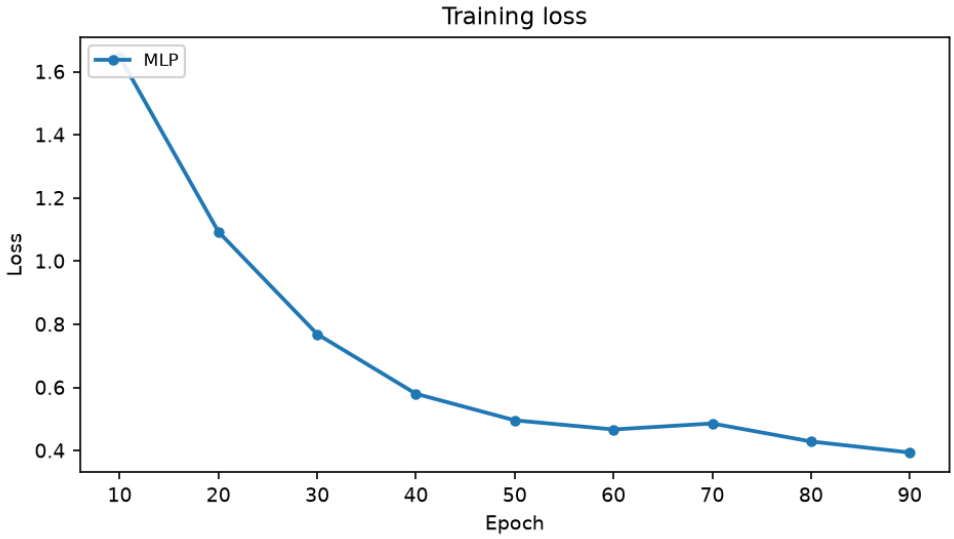


Figure 2: Training loss for the models in this experiment.

At epoch 50, the loss of MLP is over 0.4 (Figure 2). MLP has a loss of 0.856 at epoch 40. As shown in Figure 2, MLP has a loss of 0.536 at epoch 60. MLP has a loss of 0.429 at epoch 80.

Figure 3 shows the validation accuracy by dataset.

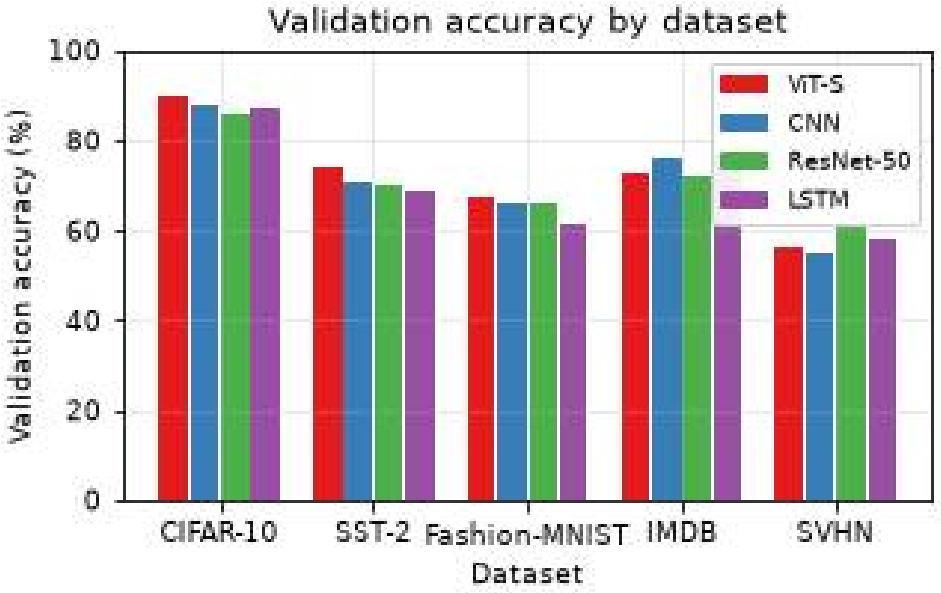


Figure 3: Validation accuracy by dataset for the models in this experiment.

As shown in Figure 3, ResNet-50 achieves 70% validation accuracy on SST-2. As shown in Figure 3, LSTM outperforms ViT-S by 1.5 percentage points on SVHN. On SVHN, CNN obtains a validation accuracy of 54.7%.

Figure 4 shows the validation accuracy during training.

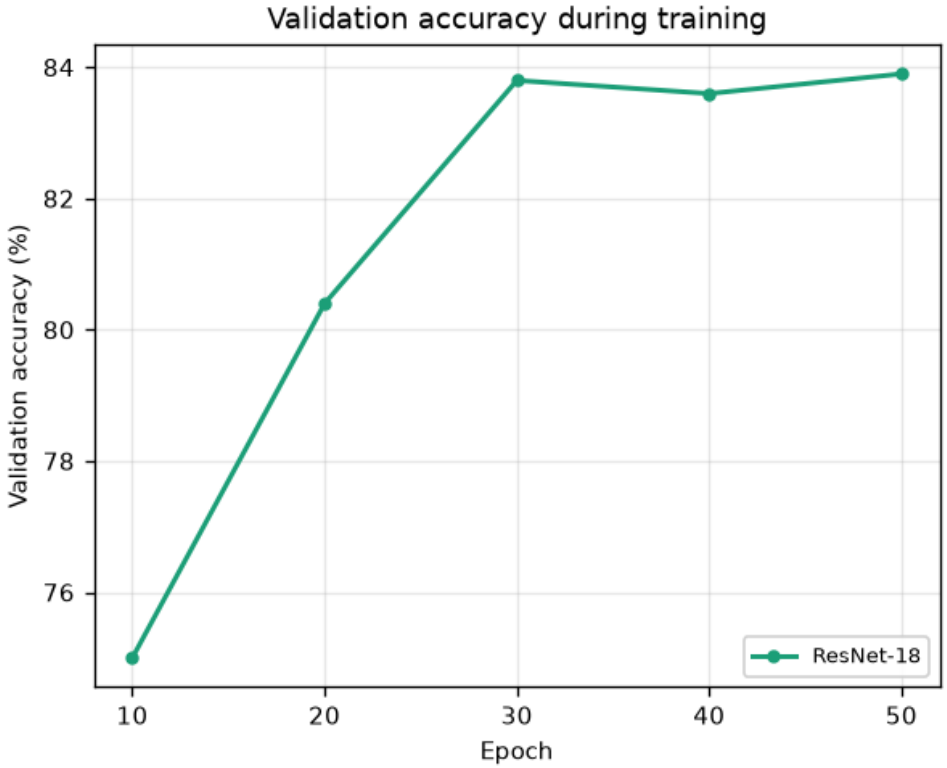


Figure 4: Validation accuracy during training for the models in this experiment.

At epoch 50, ResNet-18 achieves 83.6% validation accuracy. From epoch 10 to epoch 20, the validation accuracy of ResNet-18 improves by a relative 7.2%. ResNet-18 attains a validation accuracy of 80.4% at epoch 20. As shown in Figure 4, ResNet-18 attains a validation accuracy of over 80% at epoch 40.

3 Discussion

Overall, the results are in line with what we expected from the literature.